Quiz-2, CSE667 (DMMRS) Full marks: 20 15-04-2026

Your answers should be i) precise, ii) to-the-point, and iii) with legible handwriting.

1. Suppose a qudcopter seeks a landing zone which is marked with many bright lamps. The quadcopter has a brightness sensor with the following details:

   - Binary sensor measurement: $z \in \{\text{bright}, \neg\text{bright}\}$
   - Sensor model:

     $$p\left(z = \text{bright} \mid x = \text{home}\right) = 0.6$$
     $$p\left(z = \text{bright} \mid x = \neg\text{home}\right) = 0.3$$

   - Binary state: $x \in \{\text{home}, \neg\text{home}\}$
   - Prior: $p(x = \text{home}) = 0.5$

   Assume that the robot observes *light*. What is the probability of the robot is above its home ? In the next time step, robot observes $z = \neg\text{bright}$. What is the probability of robot is above its home ? 2+2

   Solution:

1. Robot observes z = bright. We have to calculate x_0 We can apply Bayes filter here. Note that we do not

$bel(x = home)$.

have to consider state transition, only measurement needs to be taken care of. Time point 1

$\overline{bel}(x_1 = home) = \overline{bel}(x_1 = \neg home) = 0.5$

$bel(x_1 = home) := \eta \, p(z_1 = bright \mid x_1 = home) \cdot \overline{bel}(x_1 = home)$

$= \eta \; 0.6 \times 0.5 = 0.3\eta$ — (1)

$\ldots 0.5 = 0.15\eta$

$bel(x_1 = \neg home) = \eta \; 0.3 \times 0.5 = 0.15\eta$

$0.3\eta + 0.15\eta = 1 \Rightarrow \eta = \frac{20}{9}$ — (2)

$bel(x_1 = home) = \frac{6}{9}$ [From (1)]

$bel(x_1 = \neg home) = \frac{3}{9}$

Time point 2 : [observes $z_2 = \neg$bright]

$\overline{bel}(x_2 = home) = \frac{6}{9}$ $\quad \overline{bel}(x_2 = \neg home) = \frac{3}{9}$

$bel(x_2 = home) = \eta \, p(z_2 = \neg bright \mid x_2 = home) \; \overline{bel}(x_2 = home)$

$= \eta \; 0.4 \times \frac{6}{9} = \frac{2.4}{9}\eta$

$bel(x_2 = \neg home) = \eta \, p(z_2 = \neg bright \mid x_2 = \neg home) \; \overline{bel}(x_2 = \neg home)$

$= \eta \; 0.7 \times \frac{3}{9} = \frac{2.1}{9}\eta$

$\frac{2.1}{9}\eta + \frac{2.4}{9}\eta = 1 \Rightarrow \eta = 2$

$\Rightarrow bel(x_2 = home) = \frac{8}{15}$

Evaluation:

- Correct methodology (either Bayes filter or directly Bayes theorem): 1
- Correct result: 1

2. Suppose a mobile robot is moving on a long straight road. Our location x will simply be the position along this road. Suppose that, initially, the robot is believed to be at location $x_{\text{init}} = 1,000m$, but we happen to know that this estimate is uncertain. Based on the uncertainty, we model our initial belief by a Gaussian with variance $\sigma^2_{\text{init}} = 900m^2$.

   To find out more about our location, we query a GPS receiver. The GPS tells us our location is $z_{\text{GPS}} = 1,100m$. The GPS receiver is known to have an error variance of $\sigma^2_{\text{GPS}} = 100m^2$.

   i) Write the pdfs of the prior $p(x)$ and the measurement $p(z \mid x)$. 2

   ii) What is the posterior $p(x \mid z)$? 2

Solution:

2 (i) The prior belief over position x is Gaussian.

$$\therefore \; p(x) = N(x; \underbrace{1000}_{x_{init}}, \underbrace{900}_{\sigma^2_{init}}) \quad [\text{Prior } p(x)]$$

$$p(x) = \frac{1}{\sqrt{2\pi \cdot 900}} \cdot \exp\left[-\frac{(x-1000)^2}{2 \cdot 900}\right]$$

$$\left(\text{using } p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]\right)$$

Measurement $p(z|x)$:

$$p(z|x) = N(z; \underbrace{x}_{\text{prior}}, 100)$$

Viewed as a function of x

$$p(z|x) = \frac{1}{\sqrt{2\pi \cdot 100}} \exp\left[-\frac{(z-x)^2}{2 \cdot 100}\right]$$

(ii) $p(x|z) = N(x; \mu_{post}, \sigma^2_{post})$

$$\sigma^2_{post} = \left(\frac{1}{900} + \frac{1}{100}\right)^{-1} = 90$$

$$\mu_{post} = \sigma^2_{post}\left(\frac{1000}{900} + \frac{1100}{100}\right) = 1090$$

$$p(x|z) = N(x; 1090, 90)$$

Evaluation:

- Q2i) Correct expression for prior and $p(z \mid x)$: 1+1
- Q2i) In case of incompelte answer, general expression for normal dist.: 1
- Q2ii) 2 will be given to all as it was not discussed in class.

3. Write the *Markov localization* algorithm. 3

Solution:

Algorithm Markov_localization($bel(x_{t-1}), u_t, z_t, m$)**:**

for all x_t *do*

$\overline{bel}(x_t) = \int p(x_t \mid u_t, x_{t-1}, m)\ bel(x_{t-1})\ dx_{t-1}$

$bel(x_t) = \eta\ p(z_t \mid x_t, m)\ \overline{bel}(x_t)$

endfor

return $bel(x_t)$

Table 7.1 Markov localization.

Evaluation:

- Input parameters: 0.5
- Expression of $\overline{bel}$: 1
- Expression of bel: 1
- Output: 0.5

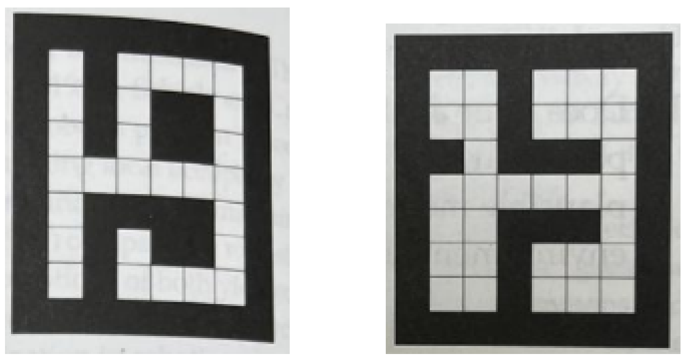
4. Consider a simplified global localization in the following grid-style environments (Figure 1). In each environment the robot will be placed at the random position facing *North*. Come up with an open-loop localization strategy that contains a sequence of the following commands:

   Action L: Turn left 90 degrees

   Action R: Turn right 90 degrees

   Action M: Move forward until you hit an obstacle

   At the end of this strategy, the robot must be at a predictable location. For each such environment, provide the *shortest* such sequence (only "M" actions count). State where the robot will be after executing your action sequence, for the two environments in Figure 1. You can number the grid cells for your convenience. 3



(a) (b)

Figure 1: Grid-style environments

Solution:

4. The robot can be placed anywhere initially, with a heading of North.
We have to choose the sequence of actions (moves) in a way that —
i) there is a unique location which can be reached.
– even if we end up at multiple indistinguishable locations from which an unique location can be reached, that should work.

Or,

ii) there are more than one distinguishable locations reachable from the possible initial locations.

In Fig (a), if we apply M, RM, RM, RM the robot will end up at either the bottom left corner cell or the bottom middle corner cell. These cells are distinguishable (criteria ii)

In Fig (b), the robot will end up at ... left or bottom

either the bottom left middle corner cell if we apply M, RM, RM, RM. These cells are indistinguishable, though the robot can reach an unique cell from those cells.

For both (a) and (b) the minimum number of "M" action is 4.

Evaluation:

- if end cells are determined (and reasoning is given): 1

- For minimum number of "M" moves: 0.5

5. In *Monte Carlo localization* the expected value of the location computed by the algorithm differs from the true expected value.

   Consider a world with four possible robot locations: $X = \{x_1, x_2, x_3, x_4\}$. Initially, we draw $N = 5$ samples uniformly from among those locations. Let Z be a Boolean sensor variable with the following conditional probabilities:

   $$p(z \mid x_1) = 0.8,\ p(z \mid x_2) = 0.4,\ p(z \mid x_3) = 0.1 \text{ and } p(z \mid x_4) = 0.1.$$

   Assume that we only generate one new sample in the resampling process, regardless of N. This sample might correspond to any of the four locations in X. Thus, the sampling process defines a probability distribution over X.

   What is the resulting probability distribution over X for this new sample ? 3

   Solution:

5. Because the initial samples ($N=5$) are drawn uniformly from the locations, so, before weighting with measurement, the distribution is uniform.

Resampling $\longrightarrow$ each particle is weighted by $p(z|x_i)$.

Since all states have equal prior, the resampling dist is proportional to the likelihoods.

$$P(x_i) \propto p(z|x_i)$$

The weights (final prob) needs to be normalized.

$$P(x_1) = \frac{0.8}{0.8+0.4+0.1+0.1} = \frac{4}{7} = \underline{0.57}$$

$$P(x_2) = \frac{0.4}{1.4} = \underline{0.28}$$

$$P(x_3) = \frac{0.1}{1.4} = \underline{0.07}$$

$$P(x_4) = \frac{0.1}{1.4} = \underline{0.08}$$

Evaluation:

- Final probability distribution over states: 1

- Final probability distribution is proportional to measurement likelihood: 1
- Final probability distribution is calculated by doing normalization: 1

6. In *Occupancy Grid mapping*, the idea is to calculate the posterior over possible maps given the data. Consider a discrete workspace with fine-grained grids. Each grid cell has a binary occupancy value. Write down the necessary equations for estimating the occupancy information using *log odds ratio* for a *static* workspace. 3

Solution:

6.

$$bel_t(x) = 1 - \frac{1}{1 + \exp(l_t)}$$

$$l_t = l_{t-1} + \underbrace{\log \frac{p(x/z_t)}{1 - p(x/z_t)}}_{\text{inverse-sensor-model}(m, x_t, z_t)} - \underbrace{\log \frac{p(x)}{1 - p(x)}}_{l_0}$$

Evaluation:

- Expression of $bel_t(x)$: 1.5
- Expression of l_t: 1.5